BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE200: Technical Writing and Presentation

DIFFUSION MODELS

A Technical Overview of Progressive Denoising for Image Generation

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This report presents a concise technical overview of diffusion models, emphasizing how progressive denoising enables high-fidelity and diverse image synthesis.

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Abstract

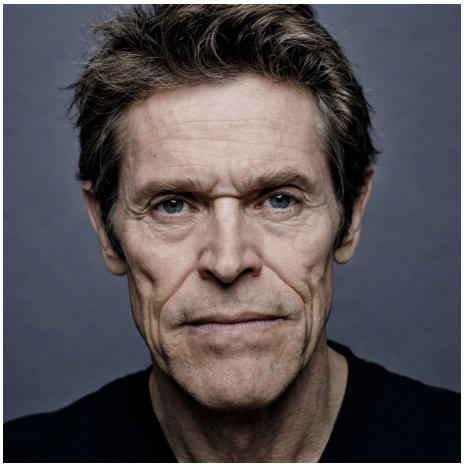
Diffusion models are generative models that learn to reverse a gradual noising process. Instead of creating an image in one step, they start from random noise and refine it through repeated denoising until meaningful structure appears. In this report, we summarize the main ideas behind Denoising Diffusion Probabilistic Models (DDPMs) [1, 2]. We cover the forward and reverse processes, the role of the U-Net backbone with timestep conditioning, and the standard noise-prediction objective used during training. We also discuss why diffusion models are now widely used in generative modeling, especially because training is comparatively stable and output quality is strong [3, 4]. Finally, we highlight practical and ethical concerns, including slow sampling, high computational cost, and responsible deployment.

1 Introduction

Generative modeling studies how to learn a probability distribution from observed data so that new samples can be drawn from it. For images, this means generating outputs that look realistic at both local and global levels, not only matching texture but also preserving object structure and consistency. Among current approaches, diffusion methods have drawn a lot of attention because they combine a clear probabilistic foundation with strong empirical results [1, 5].

Figure 1 illustrates this motivation. The generated face on the right appears plausible even though it does not correspond to a real person. A useful generative model must therefore do more than memorize training examples; it must capture the statistical regularities of the data well enough to synthesize new samples.

(a) A real face sample.



(b) A model-generated face sample.

Figure 1: Illustrative comparison between a real image and a generated image. A successful generative model should produce outputs that remain visually consistent with the underlying data distribution.

The central idea of diffusion modeling is to break generation into many small denoising

steps. During training, real data are gradually perturbed with Gaussian noise. Since this corruption process is predefined, the model can be trained to predict the added noise at different timesteps. During sampling, the process is reversed: the model starts from random noise and gradually reconstructs structure. In our understanding, this step-by-step setup makes learning more manageable than trying to predict a full image in one shot.

2 The Diffusion Process

The workflow of a diffusion model has two complementary parts: a *forward diffusion process* that destroys information and a *reverse diffusion process* that reconstructs structure. Figure 2 summarizes this idea in one diagram.

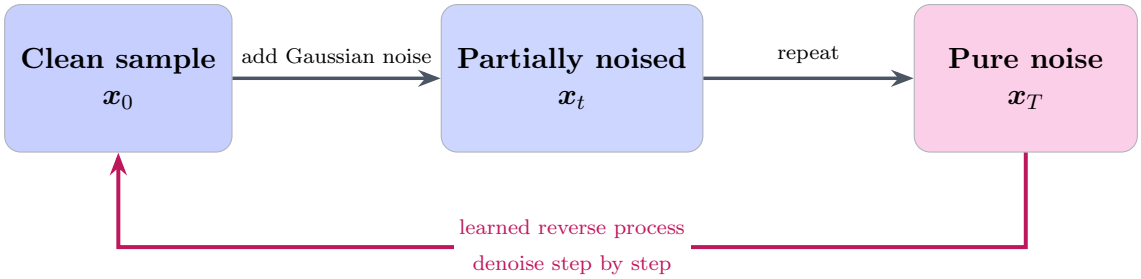


Figure 2: Conceptual overview of diffusion models. Training uses the controlled noising path, while sampling uses the learned reverse path.

2.1 Forward Diffusion Process

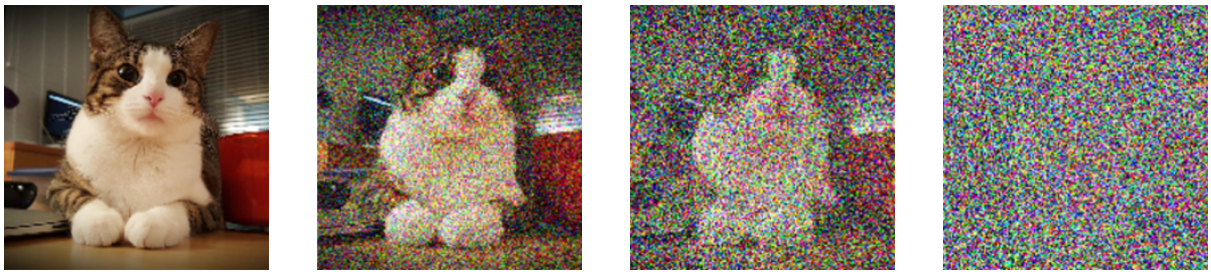
The forward process is a Markov chain that progressively perturbs a clean sample $\boldsymbol{x}_0 \in \mathbb{R}^d$ with Gaussian noise. At each timestep t, a small variance term β_t determines how much corruption is introduced:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right). \tag{1}$$

If we define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, then a sample at any timestep can be written directly as

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{2}$$

Equation 2 is convenient during training because it allows the model to sample a noisy state at any timestep directly, without simulating all previous steps.



(a) Clean (b) Low noise (c) Heavy noise (d) Pure noise

Figure 3: Visual interpretation of the forward diffusion process. A structured image is gradually transformed into an approximately Gaussian sample.

2.2 Reverse Diffusion Process

Once the data has been corrupted, the generative task becomes the reverse problem: infer how to move from $\boldsymbol{x}_t$ back to a less noisy sample $\boldsymbol{x}_{t-1}$. The model parameterized by θ learns a reverse distribution of the form

$$p_\theta(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)) \,. \tag{3}$$

In practice, many implementations train a neural network to predict the added noise rather than predict clean pixels directly. This reparameterization simplifies learning because the network only needs to estimate the perturbation that separates $\boldsymbol{x}_t$ from the underlying clean sample [1, 2].

3 Model Architecture and Training

In practice, strong performance depends not only on the probabilistic formulation, but also on the neural architecture used to model the reverse process. Most image diffusion systems employ a U-Net-style backbone with multi-scale feature extraction and skip connections [6]. The network receives two inputs: a noisy image $\boldsymbol{x}_t$ and an embedding of the timestep t.

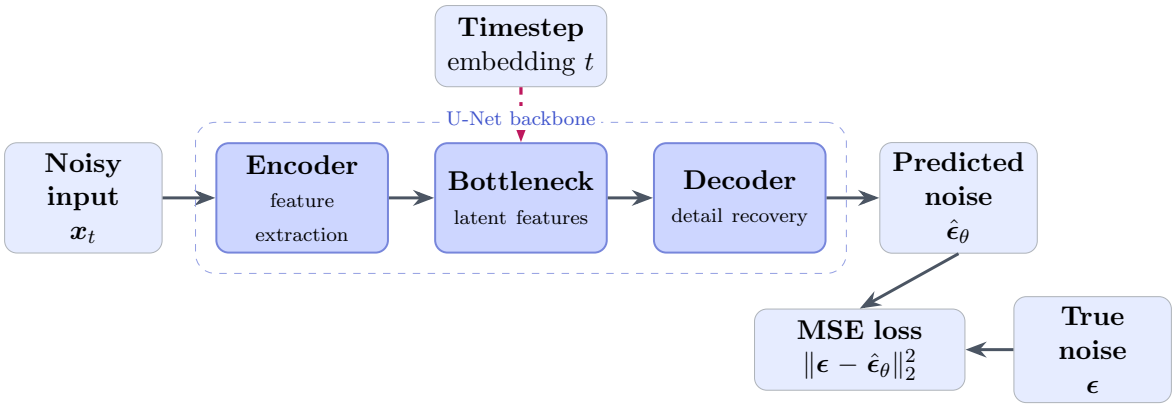


Figure 4: A simplified training view of an image diffusion model. The network receives a noisy image and the current timestep, predicts the noise component, and is trained with a mean-squared error objective.

3.1 Noise-Prediction Objective

The standard DDPM training objective is

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\boldsymbol{x}_0,t,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\|_2^2 \right] . \tag{4}$$

This loss is convenient because the target noise $\boldsymbol{\epsilon}$ is known exactly; it is sampled by the training procedure itself. As a result, supervision is explicit and comparatively stable. Instead of learning to generate a final image in one pass, the network solves many smaller denoising tasks.

3.2 Training Workflow

The complete training routine can be summarized as Algorithm 1. The procedure alternates between sampling a real image, corrupting it to a randomly chosen timestep, and updating the network so that its predicted noise matches the actual perturbation.

Algorithm 1 Training a diffusion model with noise prediction

Require: Training dataset $\mathcal{D}$, noise schedule $\{\beta_t\}_{t=1}^T$, neural network $\boldsymbol{\epsilon}_\theta$

```
while training has not converged do
    Sample a clean image x_0 ~ D
    Sample a timestep t uniformly from {1, ..., T}
    Sample Gaussian noise ε ~ N(0, I)
    Form a noisy input using Equation 2
    Predict ε̂ = ε_θ(x_t, t)
    Compute L_simple = ||ε − ε̂||_2^2
    Update θ using gradient descent
end while
```

3.3 Sampling and Guidance in Practice

At inference time, generation starts from $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and applies the learned reverse transitions repeatedly until $\boldsymbol{x}_0$ is obtained. A practical challenge is that this process can require many steps. Two common improvements are: (i) deterministic or reduced-step samplers such as DDIM, which decrease latency while preserving useful quality [5]; and (ii) guidance strategies that strengthen condition alignment, such as classifier guidance and classifier-free guidance [7, 8]. These techniques are central to modern text-to-image systems because they improve controllability and sample fidelity.

Table 1 collects the most important symbols used in the mathematical discussion so that the notation remains easy to follow.

Symbol	Meaning
$\boldsymbol{x}_0$	A clean training sample drawn from the data distribution.
$\boldsymbol{x}_t$	The noisy version of the sample at diffusion step t.
β_t	Variance schedule controlling how much noise is added at step t.
α_t	Shorthand for $1 - \beta_t$.
$\bar{\alpha}_t$	Cumulative product $\prod_{s=1}^{t} \alpha_s$, measuring the remaining signal after t steps.
$\boldsymbol{\epsilon}$	Gaussian noise used to corrupt the sample.
$\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$	Neural network prediction of the added noise.

Table 1: Notation used in the forward process, reverse process, and training objective.

4 Why Diffusion Models Perform Well

Diffusion models often perform strongly in practice for several related reasons. First, once Gaussian noise is sampled, the training target is known analytically, which avoids the adversarial optimization instability often seen in GAN training. Second, iterative denoising lets coarse structure emerge before fine details are refined. Third, conditioning inputs such as text, class labels, or masks can be integrated naturally into the denoising network, making the framework useful across many tasks.

These characteristics help explain why diffusion systems became competitive with, and in many settings preferable to, earlier generative approaches. Table 2 summarizes typical trade-offs.

Model family	Training stability	Sample quality	Sampling speed	Main comment
VAE	High	Moderate	Fast	Strong probabilistic interpretation but often blurrier outputs.
GAN	Moderate to low	High	Very fast	Can generate sharp images but training may be unstable and prone to mode collapse.
Diffusion model	High	Very high	Slow	Excellent fidelity and diversity, but generation requires many denoising steps.

Table 2: A high-level comparison of common generative model families. The table emphasizes typical tendencies rather than strict universal rules.

5 How Performance is Evaluated

In research papers, claims about image quality are usually supported by quantitative metrics in addition to visual examples. A widely used metric is the Fréchet Inception Distance (FID), which compares feature statistics of generated and real images; lower values are generally better [9]. Many papers also report Inception Score (IS) and precision-recall style analyses to separate fidelity from diversity.

For conditional systems such as text-to-image models, alignment quality is also important: the output should follow the prompt while remaining realistic. In practice, this creates a trade-off between strict condition matching and natural-looking images, so most studies report multiple metrics rather than relying on only one.

6 Applications

Diffusion models now appear in both public-facing creative tools and specialized technical workflows. In consumer applications, they are used for text-to-image synthesis, image editing, artistic rendering, and increasingly for video generation. Figure 5 shows a few widely recognized systems associated with this broader family of methods.

(a) DALL-E. (b) Midjourney. (c) Sora.

Figure 5: Examples of widely recognized systems associated with diffusion-based generation. The figure is illustrative rather than exhaustive.

Beyond creative media, diffusion models are increasingly relevant in technical domains. In medical imaging, researchers use them for denoising, reconstruction, and data aug-

mentation. In scientific design, related models are being explored for molecule generation and inverse problems. Latent diffusion models further reduce the cost of high-resolution synthesis by performing denoising in a compressed latent space rather than directly in pixel space [4].

7 Limitations and Responsible Use

Despite their strengths, diffusion models still have important limitations. The first is **slow sampling**: high-quality generation may require dozens to hundreds of denoising steps, which increases latency. The second is **high computational cost**: training large systems demands substantial GPU memory, long training times, and significant electrical power. The third is **system complexity**: practical deployments often rely on careful prompt conditioning, safety filters, and post-processing pipelines.

There are also broader ethical concerns. Diffusion models can reproduce harmful biases present in their training data, imitate artistic styles without consent, or be used to create misleading synthetic media. For this reason, technical progress should be considered alongside dataset transparency, usage policies, and clear attribution practices.

8 Conclusion

Diffusion models have become one of the most practical and effective frameworks in modern generative modeling. By learning to reverse a controlled corruption process, they combine probabilistic ideas with neural network-based denoising in a way that is both mathematically grounded and useful in practice. This helps explain their strong performance in image generation and related tasks.

At the same time, understanding diffusion models is easier when the topic is presented step by step: forward process, reverse process, architecture, training objective, applications, and limitations. Framing the report this way shows both how these models generate realistic samples and where their current limitations still remain.

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